

Stage de fin d'étude:

Évaluation et développement de méthodes Deep Learning pour l'analyse de données single-cell RNA-seq.

4 janvier - 30 juin 2026

Encadrants : Victoria Bourgeais, Loann Giovannangeli, Patricia Thébault

Tutrice : Adelaïde Raguin

LaBRI

université
de **BORDEAUX**

Introduction

Identifiant Cellule	Matrice d'expression (X)			Métadonnées des cellules (obs)	
	Gène A	Gène B	Gène C	Batch	Type cellulaire
Cellule_1	0	14	0	Batch_A	Cellule Beta
Cellule_2	3	0	1	Batch_A	Cellule Alpha
Cellule_3	0	0	0	Batch_B	Cellule de Schwann
Cellule_4	24	1	0	Batch_B	Cellule Beta
Cellule_5	0	0	8	Batch_C	Lymphocyte T

Précision à l'échelle de la cellule → Identification de populations cellulaires précises

Bruitées : haute dimension, creuses + effet dropout, effet batch

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**Souvent
manquant**

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- Filtre cellules / gènes
- Normalisation
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3. Clustering

- Méthode basée sur les graphes
- Méthode basée sur la densité
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CHALLENGE

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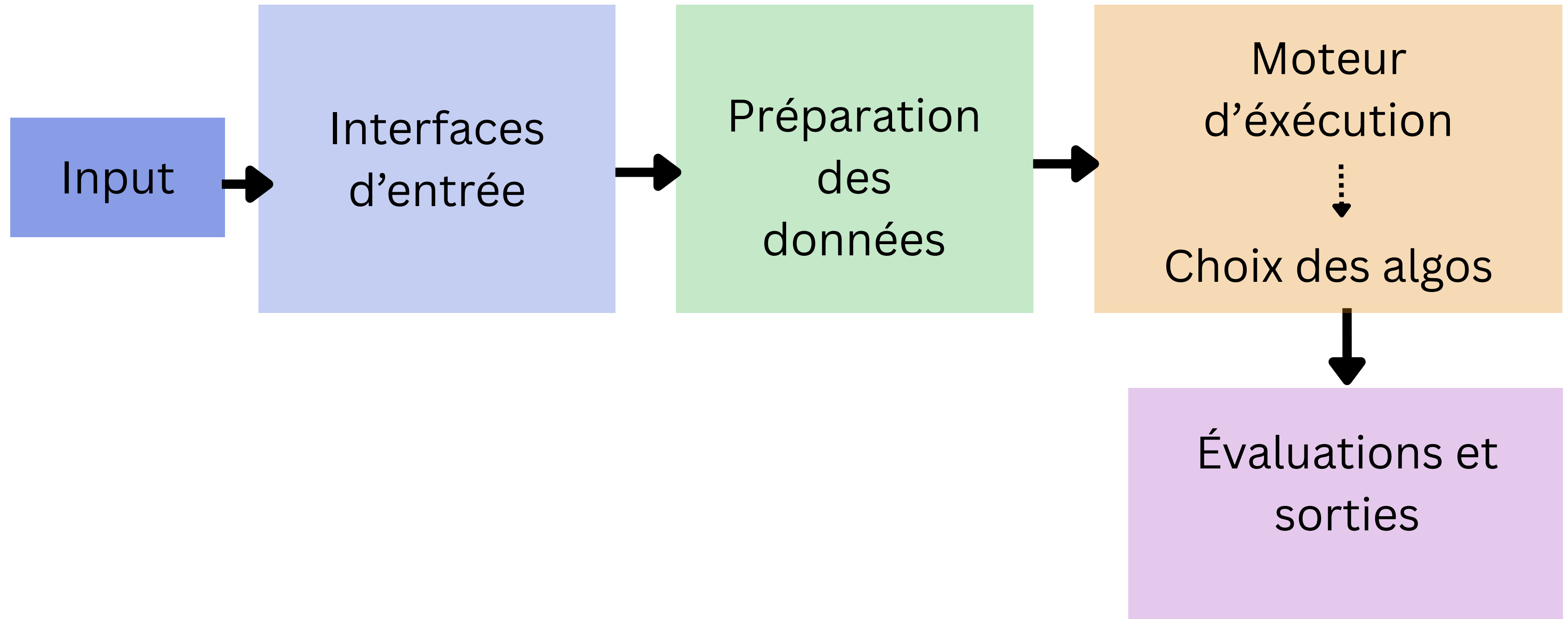
Famille	Méthode	Rôle	Framework
Approches classiques	PCA [20]	Réduction de dimension	Projection linéaire
	K-Means [10]	Clustering	Centroïdes
	Louvain [11]	Clustering	Graphe de voisinage
	Leiden [12]	Clustering	Graphe de voisinage
	HDBSCAN [13]	Clustering	Densité hiérarchique

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	scMAE [14]	Réduction de dimension	Masked AE
	DESC [15]	Les deux	AE + clustering
	scDeepCluster [16]	Les deux	AE ZINB
	scNAME [17]	Les deux	Self-supervised
	scCDCG [21]	Les deux	AE + graph embedding

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Famille	Méthode	Rôle	Framework
Cellules rares	GiniClust [22]	Clustering	Indice de Gini
	CellSIUS [23]	Clustering	Signatures locales
	scAIDE [24]	Les deux	Autoencodeur
	DeepScena [25]	Les deux	Self-supervised
	scCAD [26]	Clustering	Détection d'anomalies

Développement d'une interface graphique pour l'étude de l'état de l'art



Évaluation des frameworks de l'état de l'art : SCRBenchmark

Quick Start ⓘ

Active mode: Quick Start

Progression Workflow: 1/5

Main Workflow

• ✓ Step 1: Data Upload

Step 2: Data Split

Step 3: Preprocessing

Step 4: Algorithm Config

Step 5: Run Analysis

Standalone Tools

Clustering Visualization

Latent Re-clustering

Customize Benchmark

Results Explorer

Hyperparam Search

Documentation

Status

SCRBenchmark Platform

Single-Cell RNA-seq Clustering Benchmark Suite

Data Loading

Upload File

Reference Datasets

Upload Data File

Choose a file

Upload

10GB per file • H5, H5AD, CSV, TSV, TXT

Or Load from Path

File path

/path/to/your/data.h5ad

Load Uploaded File

Load from Path

Dataset Overview

Cells

2,730

Genes

3,451

Batches

1

Data Type ⓘ

Raw Counts (Integers)

Median Genes/Cell ⓘ

872

Mean Reads/Cell ⓘ

2,228

Sparsity ⓘ

74.7%

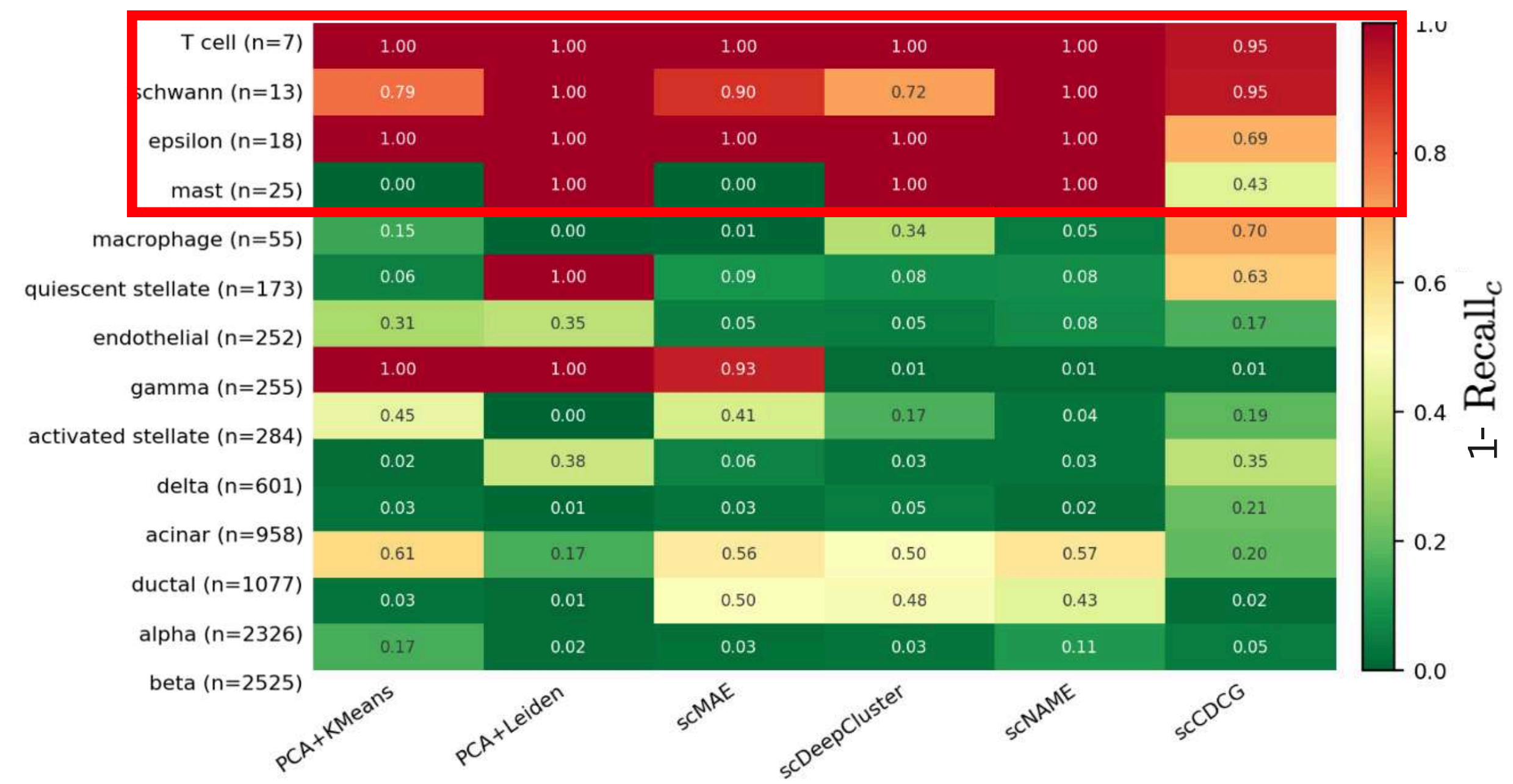
Embeddings ⓘ

0

Difficulté des frameworks à identifier les cellules rares



Erreurs en bordures de clusters



$$\text{Recall}_c = \frac{\text{cellules de la classe } c \text{ correctement retrouvées}}{\text{nombre total de cellules de la classe } c}$$

Difficulté des frameworks à corriger l'effet batch

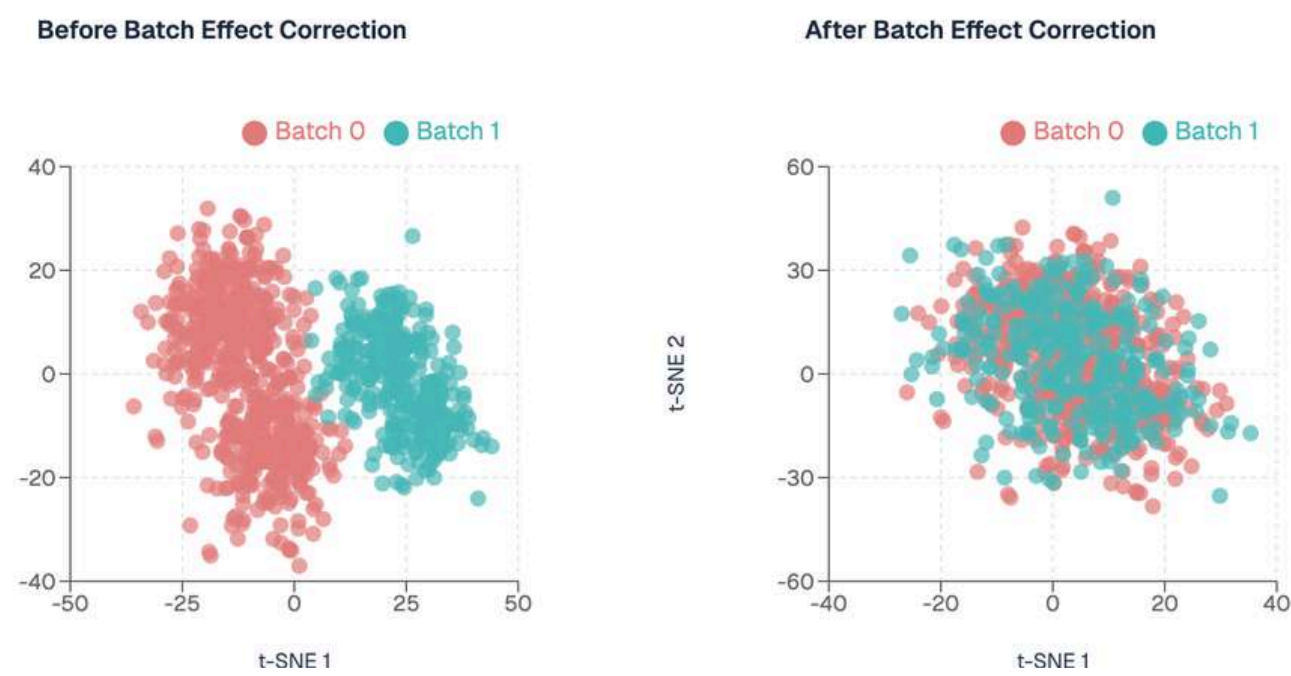


Umap de scMAE coloré par type batch d'expérience sur les données Baron Human Pancreas

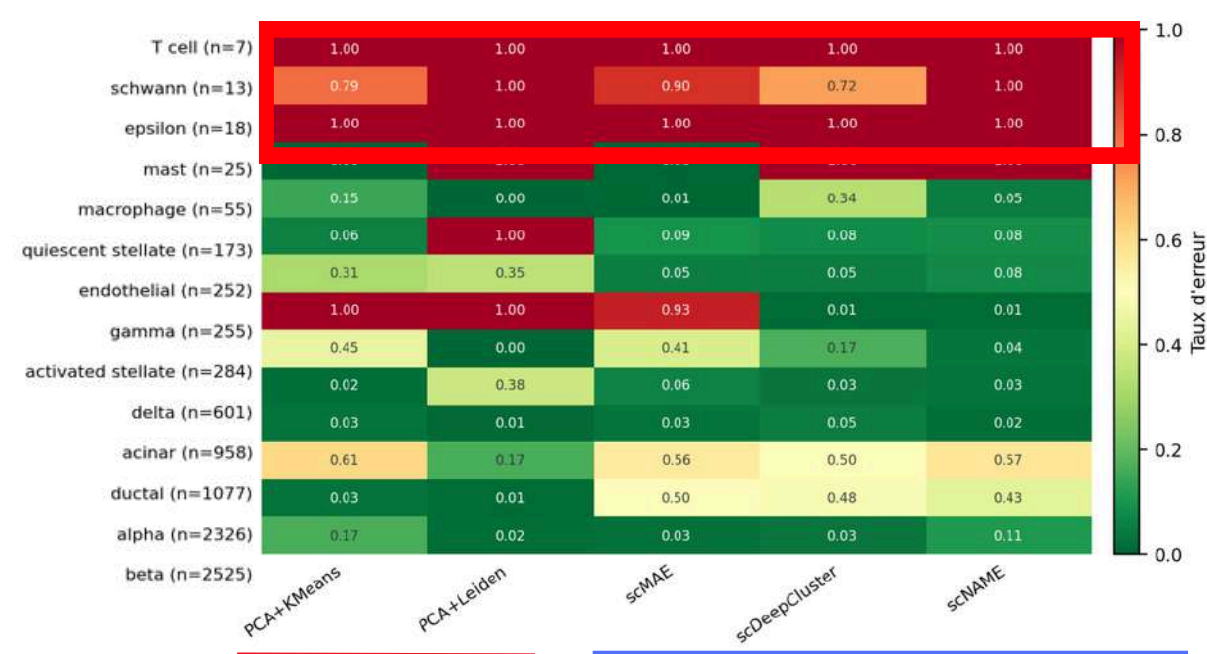
Développement d'un algorithme pour
répondre aux limites de l'état de l'art

Conclusion : Limitation des méthodes de l'état de l'art

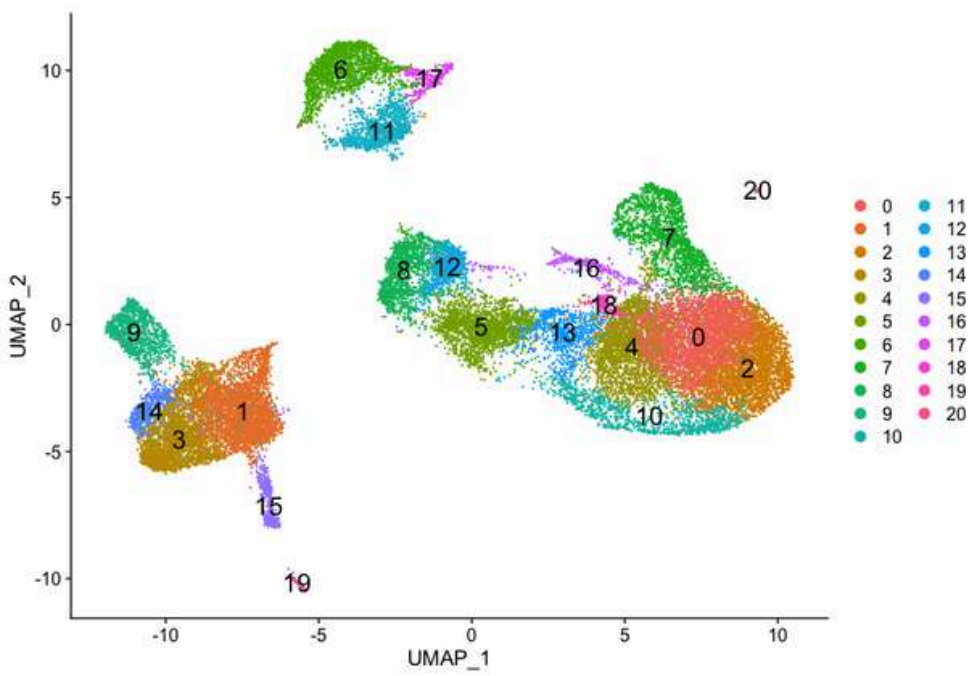
Difficultés pour faire à la fois :



Correction du batch effect



Identification des populations rares



Bon clustering global

scRAW: Representation learning for rare cell population identification

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Keywords scRNA-seq, unsupervised clustering, rare cell populations, representation learning, batch correction, imbalanced labels

Abstract Single-cell RNA sequencing (scRNA-seq) data is essential for describing tissue heterogeneity, but rare cell populations remain difficult to identify. This challenge affects learning-based clustering methods because single-cell data are high-dimensional, sparse, noisy, imbalanced, and often affected by batch structure. In deep learning models, optimization is largely driven by abundant cell types, such that rare cells often contribute too weakly to organize the latent space. To address this, we present scRAW, a rare-population autoencoder framework for scRNA-seq clustering. scRAW combines standard preprocessing with a two-phase deep learning-based training strategy. In the first phase, all cells contribute equally to the reconstruction objective. In the second phase, cell-specific weights are computed from latent pseudo-cluster size and local density such that small cell groups contribute more strongly to training. A triplet-based regularization then increases the saliency of high-weight cells in the latent space, while adversarial batch correction can be added when batch labels are available. The learned representation is finally clustered with a state-of-the-art algorithm. We evaluated our approach on a human pancreas benchmark, where scRAW reached mean ARI = 0.9005 ± 0.0314 and mean ACC = 0.9087 ± 0.0215 across 15 independent evaluations. scRAW also ranks first on several label-agreement, rare-population, internal-structure, and batch-effect metrics. These results support custom rare-cell reweighting as an effective framework-level strategy that improves rare-cell identification without degrading overall clustering quality.

1 Introduction

scRNA-seq is used to study tissue heterogeneity, and clustering is a common way to identify cell populations in these data. Clustering scRNA-seq data is challenging because the gene expression matrix is high-dimensional, sparse, noisy, and often affected by batch effects [1]. Performance also depends on preprocessing, representation learning, and partitioning choices, leading to variability across datasets and methods [9, 10, 11, 12]. Many scRNA-seq clustering approaches still rely on conventional dimensionality-reduction and clustering pipelines. In particular, PCA representations are often combined with widely used clustering algorithms such as KMeans [17], and PCA-based strategies have also been explored in methods such as pcaReduce [18]. Deep learning methods [6] follow the same general goal, but try to learn a latent space in which clustering becomes easier. In other words, the representation is optimized such that cells with similar profiles are brought closer together, while different populations are better separated before the final clustering step. These methods can then rely on different clustering strategies in the learned space, including KMeans-based objectives, graph-based clustering, or neighborhood-based contrastive clustering. Examples include scDeepCluster [19], scMAE [24], and scNAME [31]. This framing is consistent with the deep clustering literature, in which deep neural networks are used to learn clustering-friendly representations from complex data [6].

However, in standard unsupervised training, the objective function is dominated by common cell types. As a result, neural networks tend to reach a local optimum optimized for predominant samples, leaving rare populations with a small influence on the learned representation. Consequently, rare cells are often absorbed into larger clusters and become difficult to identify, despite their biological significance [21]. While non-deep learning methods such as GiniClust [22] and CellSIUS [23] were developed to improve rare-cell detection in single-cell data, they mainly operate as rare-population detection or post-clustering refinement tools. GiniClust is limited in preserving the global cluster structure, whereas CellSIUS depends on the quality of an initial fixed clustering.

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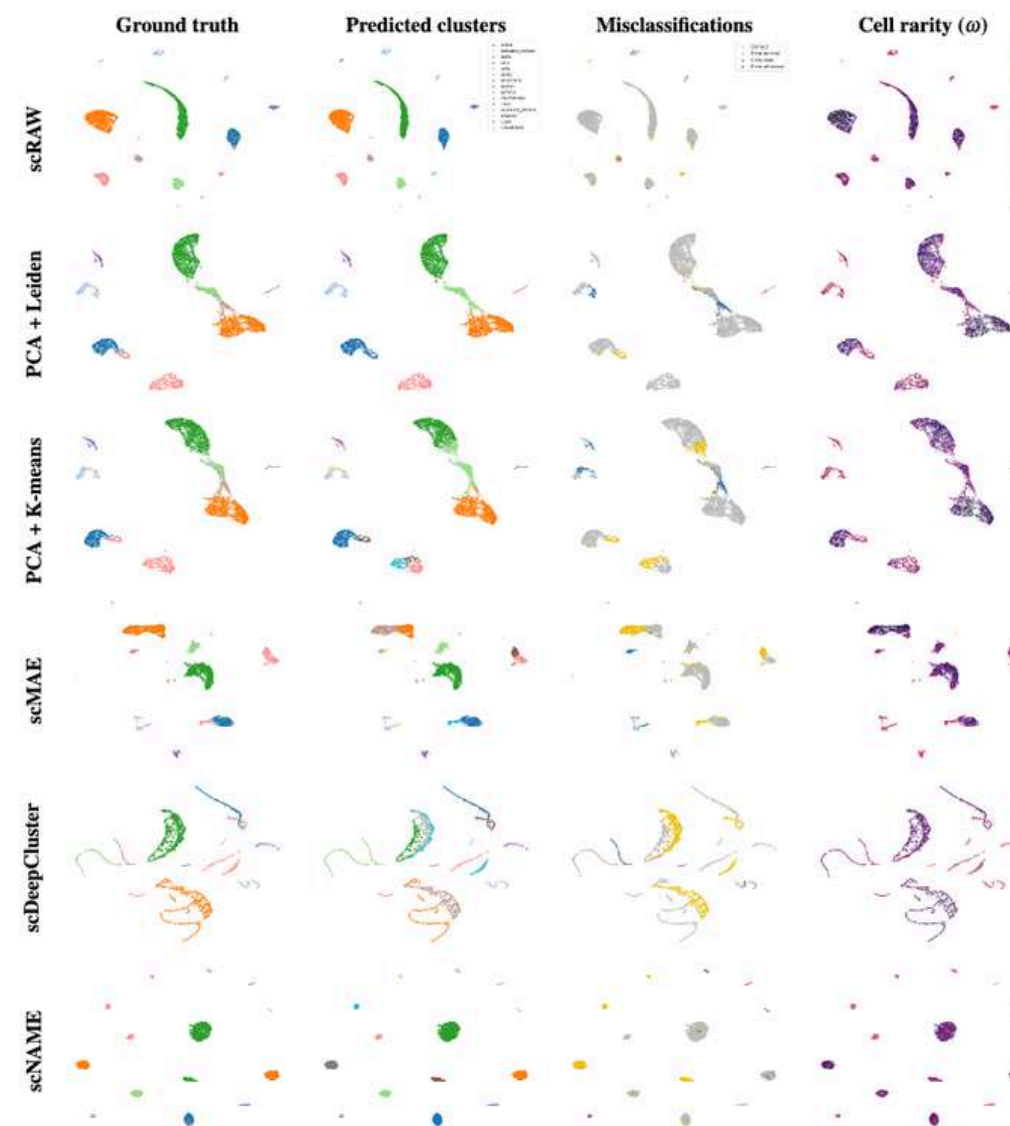


Figure 1: Representations of the latent spaces learned by all compared methods in 2D space (UMAP projection). For each method, the first two columns report ground-truth and predicted labels after Hungarian alignment. The third column highlights classification outcome at the single-cell level: correctly classified cells are shown in gray, whereas errors are colored according to the rarity of the true population. The rightmost column displays the per-cell rarity score ω (see Equation 5); for baseline methods, these values correspond to scRAW-derived weights reprojected onto the same 2D coordinates. Cells with larger ω values are shown with higher opacity.

scDeepCluster is clearly lower. This indicates that the learned scRAW representation provides the best overall trade-off between batch correction and biological conservation among the methods considered here.

Table 1. Comparison between scRAW and other algorithms on the Baron 2000-HVG benchmark. All rows report mean $\pm$ standard deviation across 15 seed reruns. For PCA+Leiden and PCA+KMeans, this dispersion mainly reflects seed-dependent variation in the final clustering stage, that is, graph construction and Leiden partitioning for PCA+Leiden, and centroid initialization for PCA+KMeans, rather than a change in the PCA preprocessing itself. Best values are shown in bold and second-best values are underlined. For scRAW, the reported results correspond to the Optuna-selected hyperparameter setting used in this study, whereas the comparison methods were evaluated with their standard or default settings and were not re-tuned specifically on Baron. UltraRareACC was recomputed from class-wise recalls as the subset accuracy over cell types with at most 30 evaluated cells.

Method	Global label-agreement metrics				
	ARI	NMI	ACC	Mean F1	BalancedACC
scRAW	0.9005 $\pm$ 0.0314	0.8788 $\pm$ 0.0162	0.9087 $\pm$ 0.0215	0.7115 $\pm$ 0.0625	0.7727 $\pm$ 0.0715
PCA+Leiden	0.8448 $\pm$ 0.0743	0.8443 $\pm$ 0.0351	0.8627 $\pm$ 0.0468	0.4310 $\pm$ 0.0758	0.4864 $\pm$ 0.0700
PCA+KMeans	0.7870 $\pm$ 0.0062	0.8036 $\pm$ 0.0032	0.8032 $\pm$ 0.0051	0.5817 $\pm$ 0.0076	0.6008 $\pm$ 0.0216
scMAE	0.7782 $\pm$ 0.0579	0.8202 $\pm$ 0.0163	0.7487 $\pm$ 0.0474	0.6262 $\pm$ 0.0077	0.6278 $\pm$ 0.0271
scDeepCluster	0.7425 $\pm$ 0.0817	0.8189 $\pm$ 0.0357	0.7521 $\pm$ 0.0563	0.5890 $\pm$ 0.0429	0.6156 $\pm$ 0.0362
scNAME	0.7441 $\pm$ 0.0505	0.8379 $\pm$ 0.0171	0.7452 $\pm$ 0.0326	0.6112 $\pm$ 0.0258	0.6236 $\pm$ 0.0305

Method	Rare-population, internal-structure, and batch-effect metrics			
	RareACC	UltraRareACC	Silhouette	scIB-E
scRAW	0.7840 $\pm$ 0.1245	0.6138 $\pm$ 0.1479	0.7157 $\pm$ 0.0602	0.5364 $\pm$ 0.0238
PCA+Leiden	0.4341 $\pm$ 0.1179	0.0000 $\pm$ 0.0000	0.4472 $\pm$ 0.0155	0.4405 $\pm$ 0.0002
PCA+KMeans	0.5106 $\pm$ 0.0342	<u>0.4540 $\pm$ 0.0552</u>	0.4489 $\pm$ 0.0012	0.4405 $\pm$ 0.0002
scMAE	0.6539 $\pm$ 0.0480	0.4476 $\pm$ 0.0776	0.3721 $\pm$ 0.0105	0.4398 $\pm$ 0.0074
scDeepCluster	0.8654 $\pm$ 0.0447	0.0593 $\pm$ 0.0786	0.1409 $\pm$ 0.0419	0.3762 $\pm$ 0.0067
scNAME	0.8435 $\pm$ 0.0714	0.0688 $\pm$ 0.0729	0.9623 $\pm$ 0.0057	0.4419 $\pm$ 0.0123

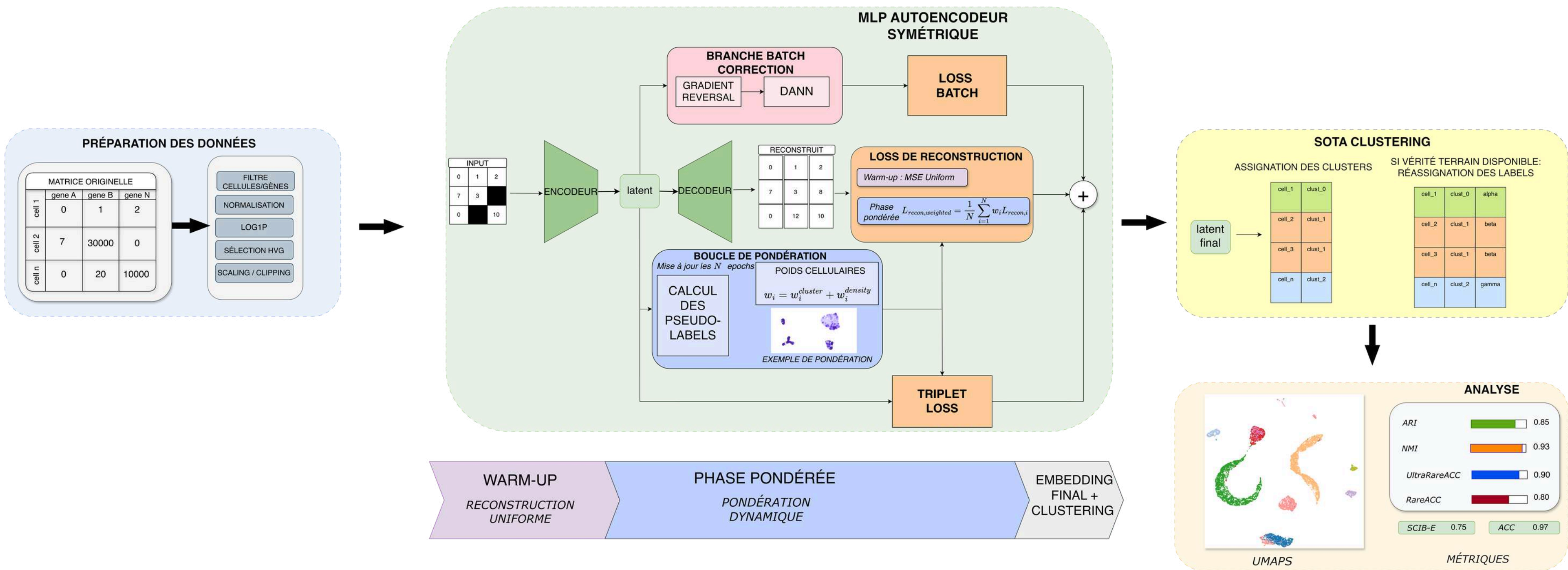
3.2 Rare Population Detection

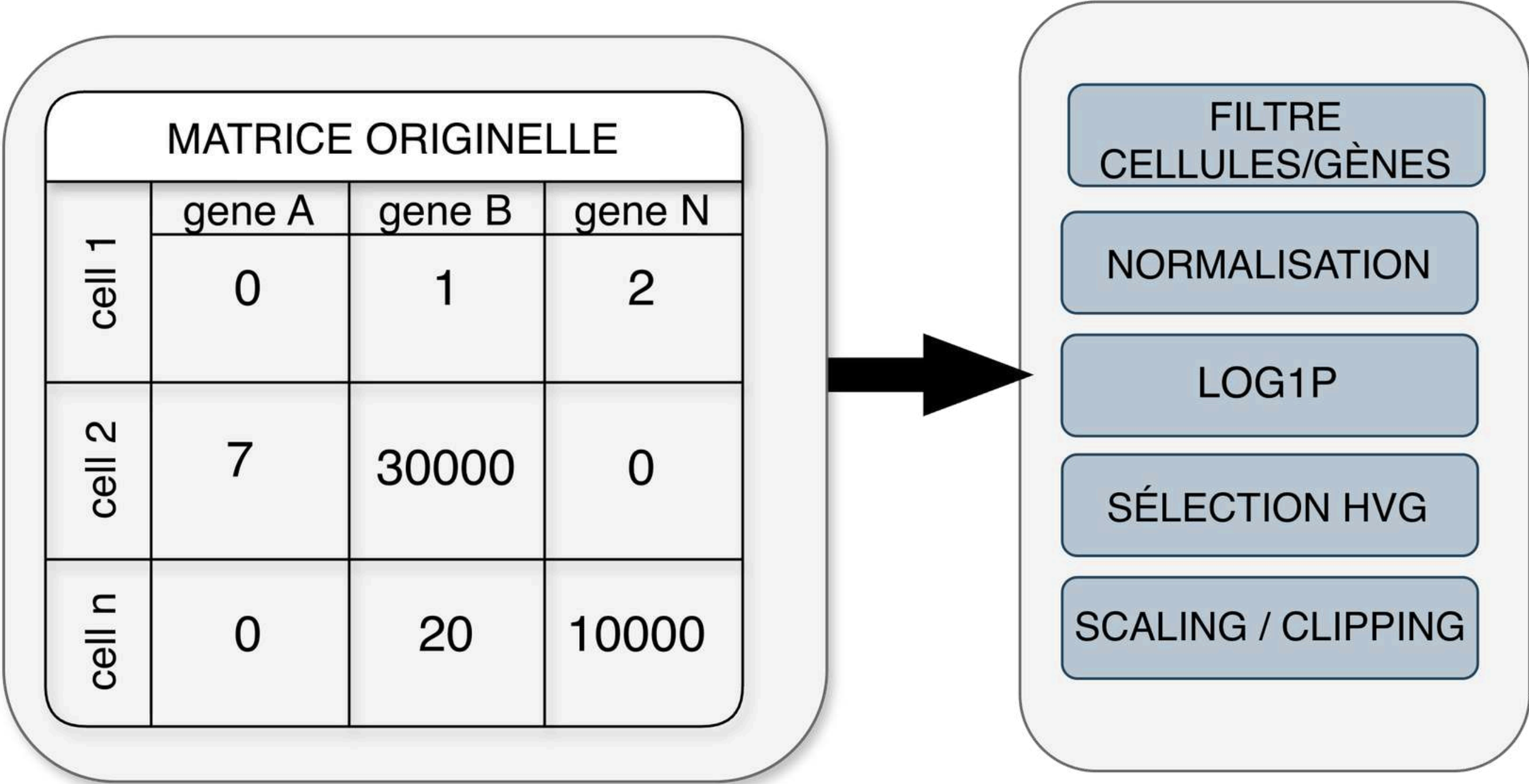
Table 2 reports the per-cell-type accuracy, defined here as class-wise recall, for the best-performing evaluation of each algorithm. The representative evaluation for each method was selected by BalancedACC across the 15 seed reruns. Among the baselines, identification is more partial and fragmented: PCA+KMeans and scMAE identify *mast* cells well, scMAE and scNAME identify the *t_cell* group in their selected runs, PCA+KMeans also identifies a large part of the *schwann* population, and scDeepCluster fully identifies *gamma* while still capturing only part of *schwann*. None of these baselines matches the overall coverage of scRAW across the smallest groups. On the broader set of rare populations, the pattern is more heterogeneous, with different baselines performing better on different cell types. In the best-performing scRAW evaluation, the only remaining extra cluster is limited in size and appears to correspond to a local split of the ductal compartment rather than to an additional distinct population. It contains 171 cells, mostly ductal (156 cells), with a small number of acinar (13) and beta (2) cells. Under Hungarian matching, predicted clusters are assigned to reference labels in a one-to-one manner. This small mixed cluster remains unmatched because the ductal label is already assigned to a larger predicted cluster, and the remaining cells do not match another reference label clearly enough. This reflects a limitation of the Hungarian algorithm, which favors a single best one-to-one correspondence even when a biological population is split into two predicted clusters. This suggests a local over-segmentation of one region rather than a systematic failure across cell types.

Figure 1 shows the clustering results for all compared methods. In scRAW, the predicted clusters match the ground-truth labels well, and the misclassification panel shows that errors are few and mostly

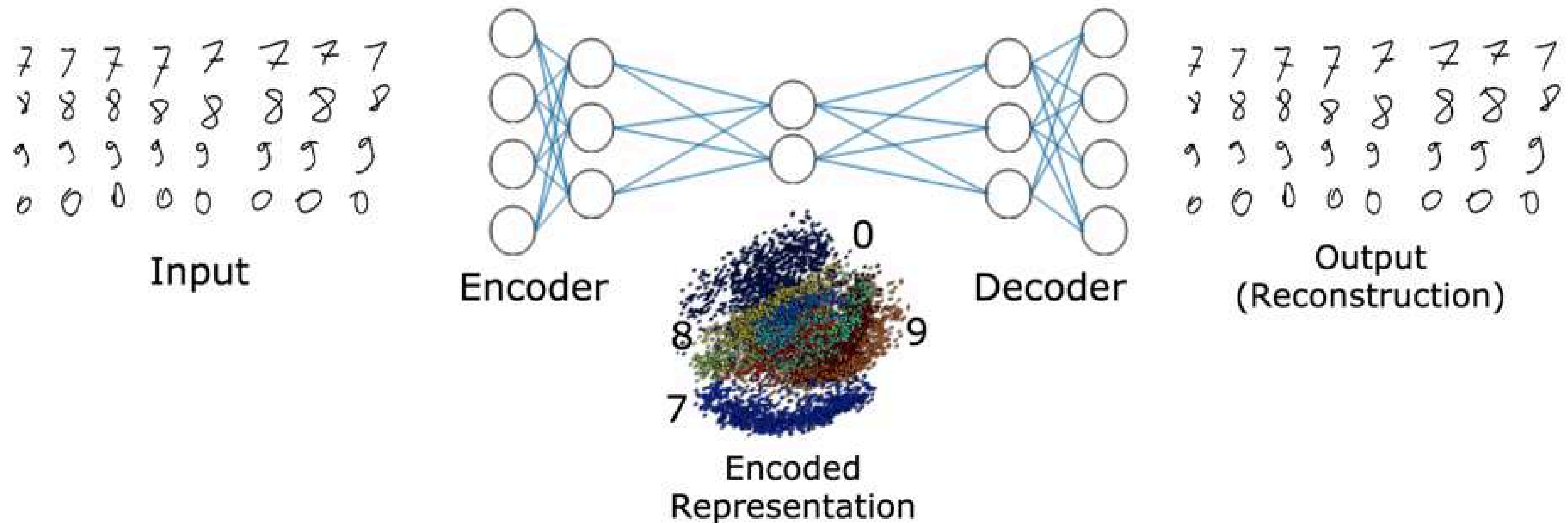
Conception de notre modèle : scRAW

ScRAW : single cell Rare Autoencoder Weighting



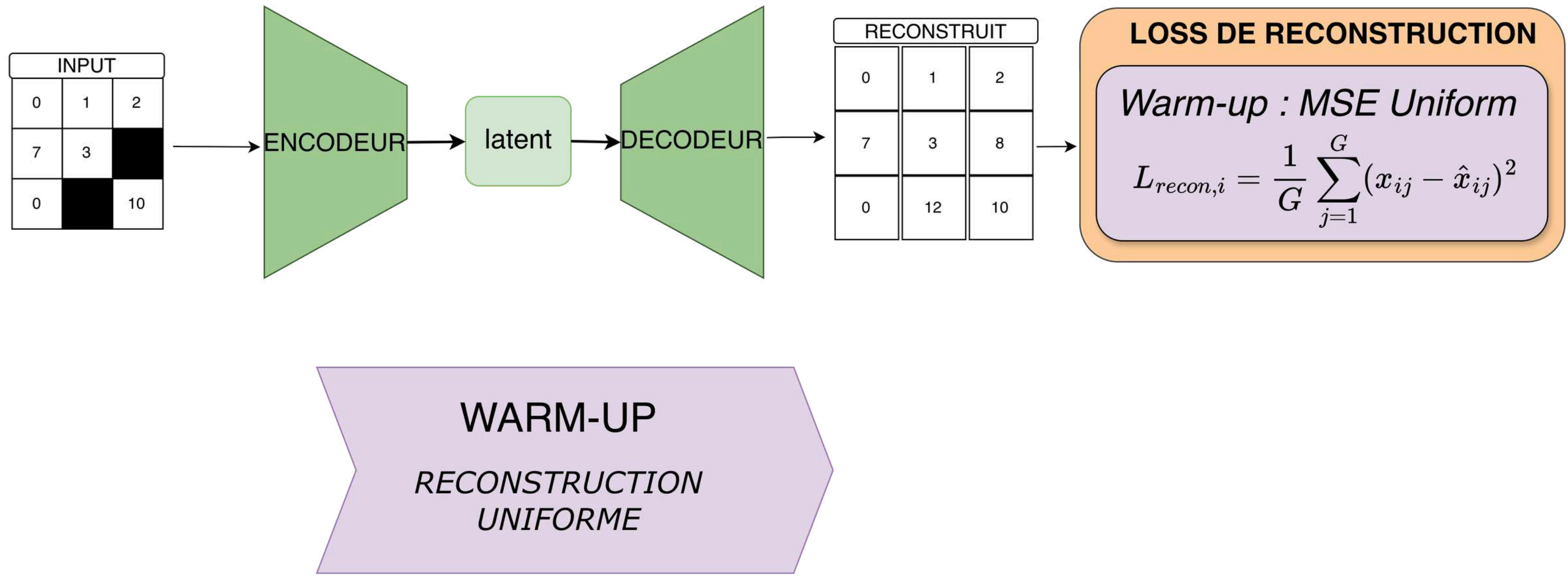


Définition : Autoencodeur



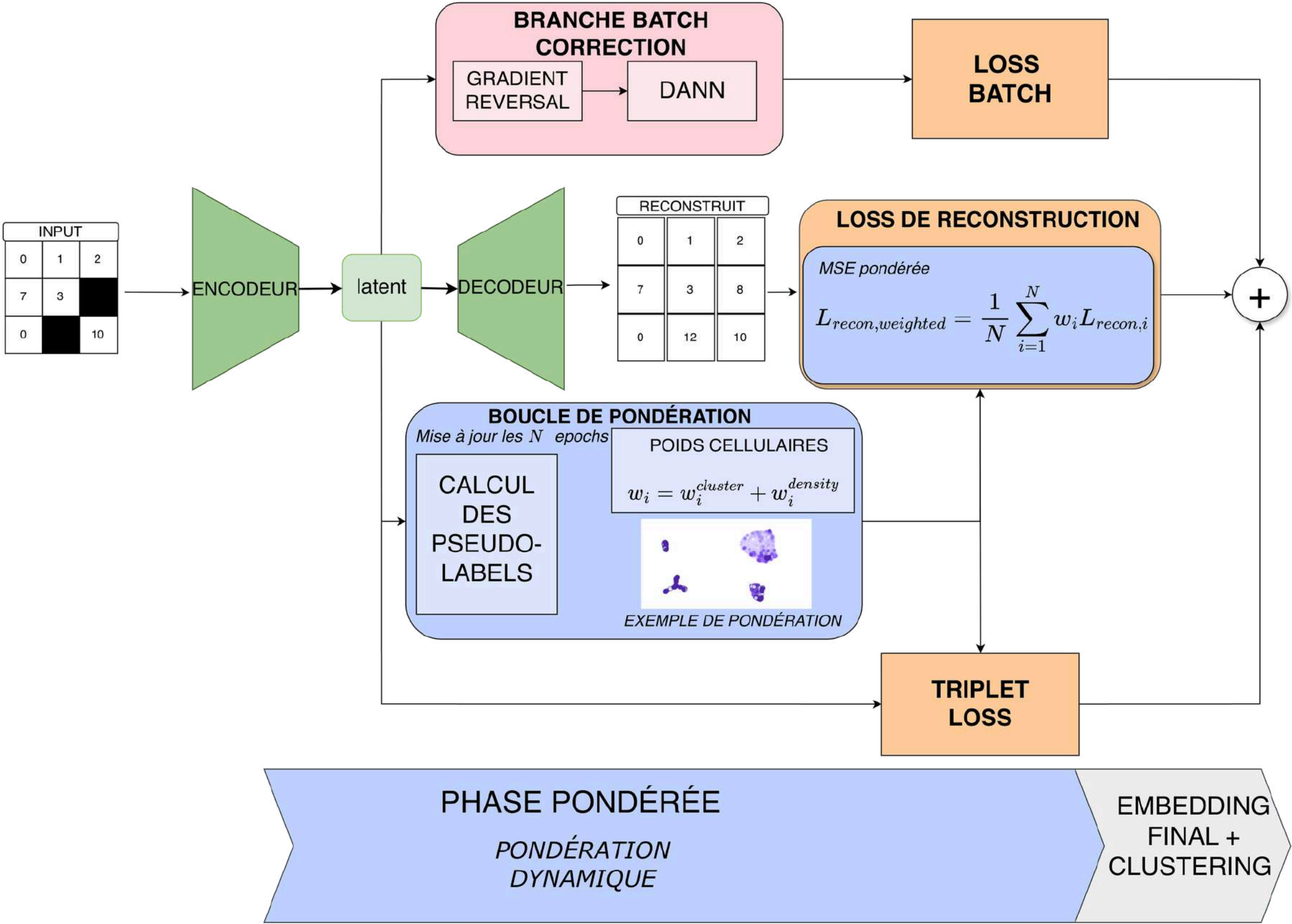
Apprentissage en minimisant ses erreurs : **Loss**

Phase 1 (WARM-UP): Autoencodeur simple avec une loss de reconstruction standard (MSE)



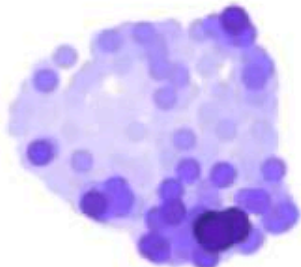
Phase 2 pondérée:

- **MSE pondérée** positivement par les cellules rares
- **Triplet Loss**
Kilian Weinberger et al (2005)
- **DANN** (Batch Correction Loss)
Ganin, Y. et al 2015.



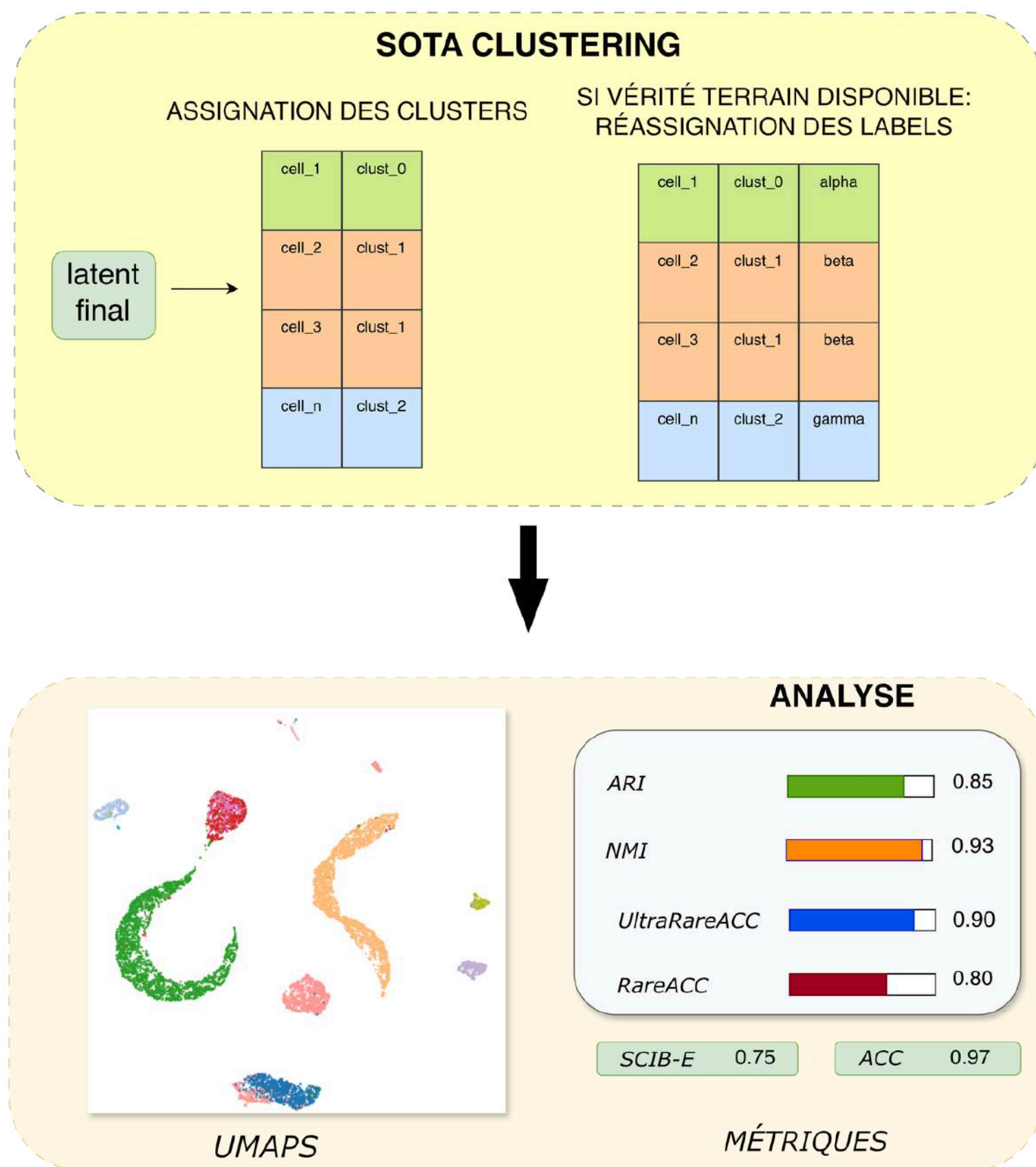
Reconstruction pondérée:

$$\mathcal{L}_{\text{recon,ponderee}} = \frac{1}{N} \sum_{i=1}^N \tilde{\omega}_i \mathcal{L}_{\text{recon},i} = \text{Reconstruction avec poids proportionnel à la rareté cellulaire}$$

$$\tilde{\omega}_i = \omega_i^{\text{cluster}} \times \omega_i^{\text{densite}} =$$
 $\omega_i^{\text{densite}} = (\text{distance } 15^{\text{ème}} \text{ voisin}) / \text{dist_médiane}$



$$\omega_i^{\text{cluster}} = 1 / \text{taille du cluster}$$

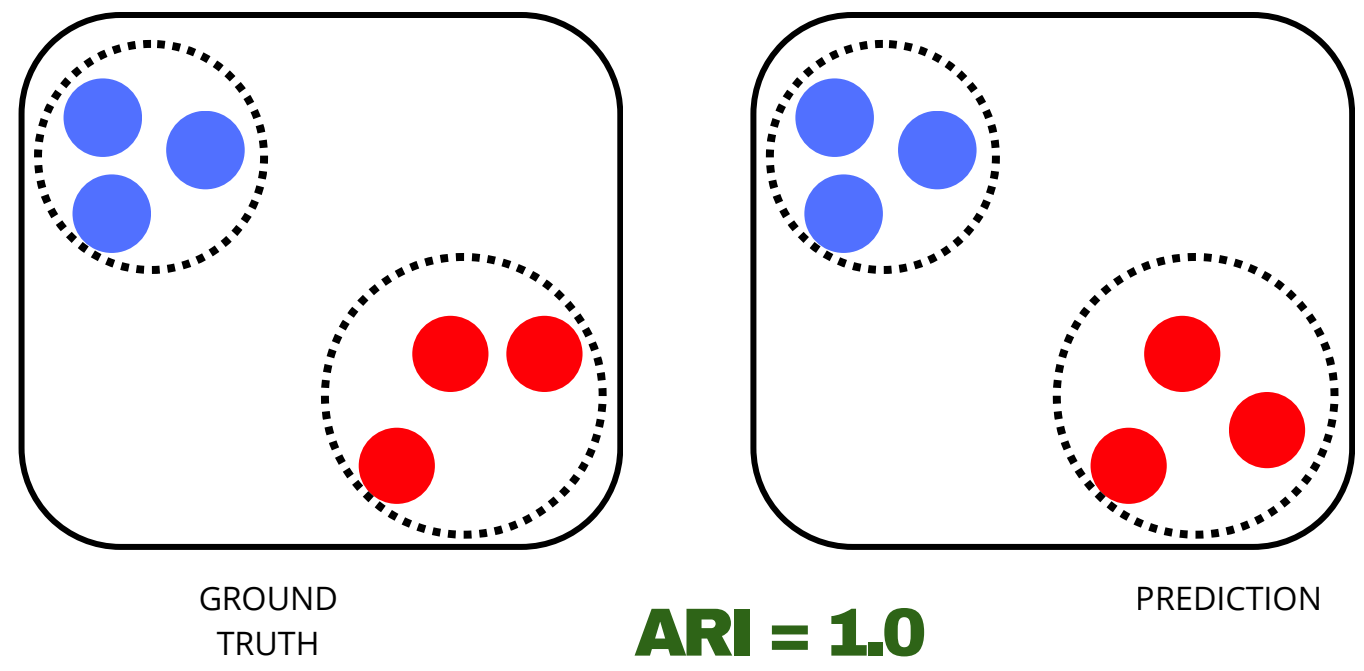


Clustering HDBSCAN

- Capable de détecter des clusters de densité très différente
- Détecte le bruit

Métrique de clustering

$$ARI = \frac{\text{Index} - \text{ExpectedIndex}}{\text{MaxIndex} - \text{ExpectedIndex}}$$



Métrique de reconnaissance des types cellulaires

$$\text{BalancedAccuracy} = \frac{1}{K} \sum_{k=1}^K \frac{TP_k}{TP_k + FN_k}$$

$$\text{RareACC} = \frac{\text{Correct predictions on rare cells}}{\text{Total number of rare cells}}$$

$$\text{UltraRareACC} = \frac{\text{Correct predictions on ultrarare cells}}{\text{Total number of rare cells}}$$

Métrique de l'effet batch

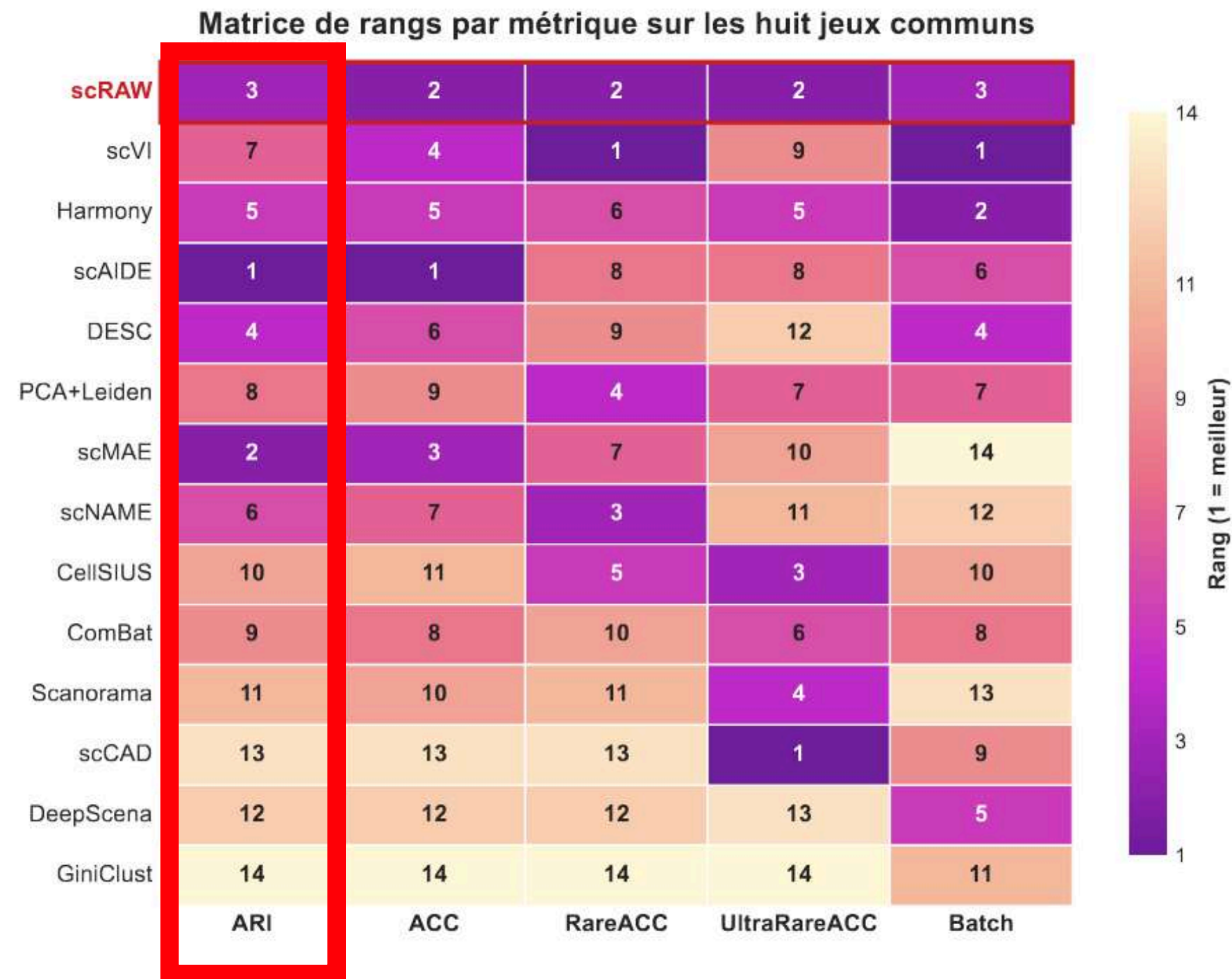
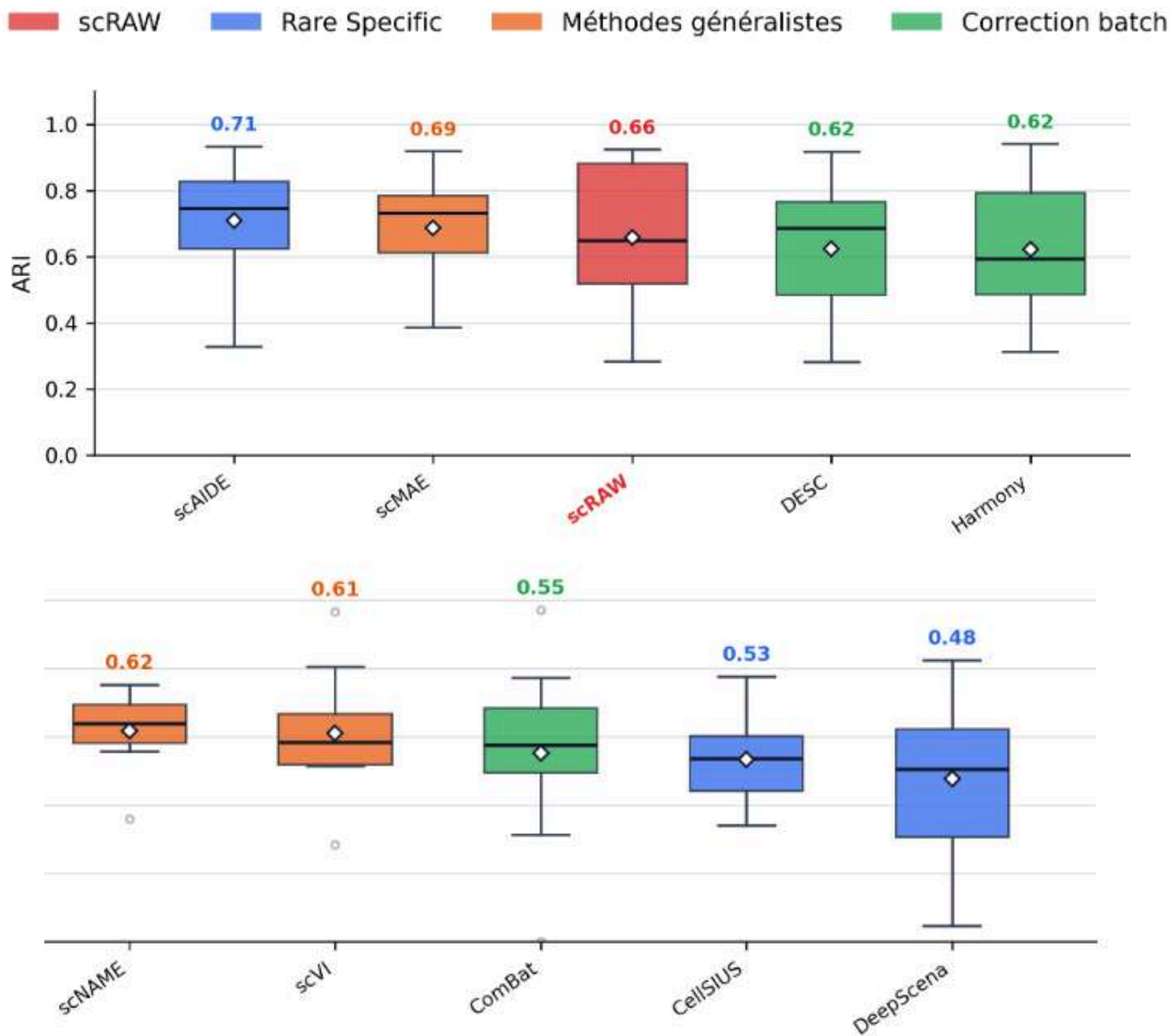
scIB-E batch $\approx$
mean(Silhouette
batch, iLISI, KBET,
Graph connectivity),

Datasets étudiés

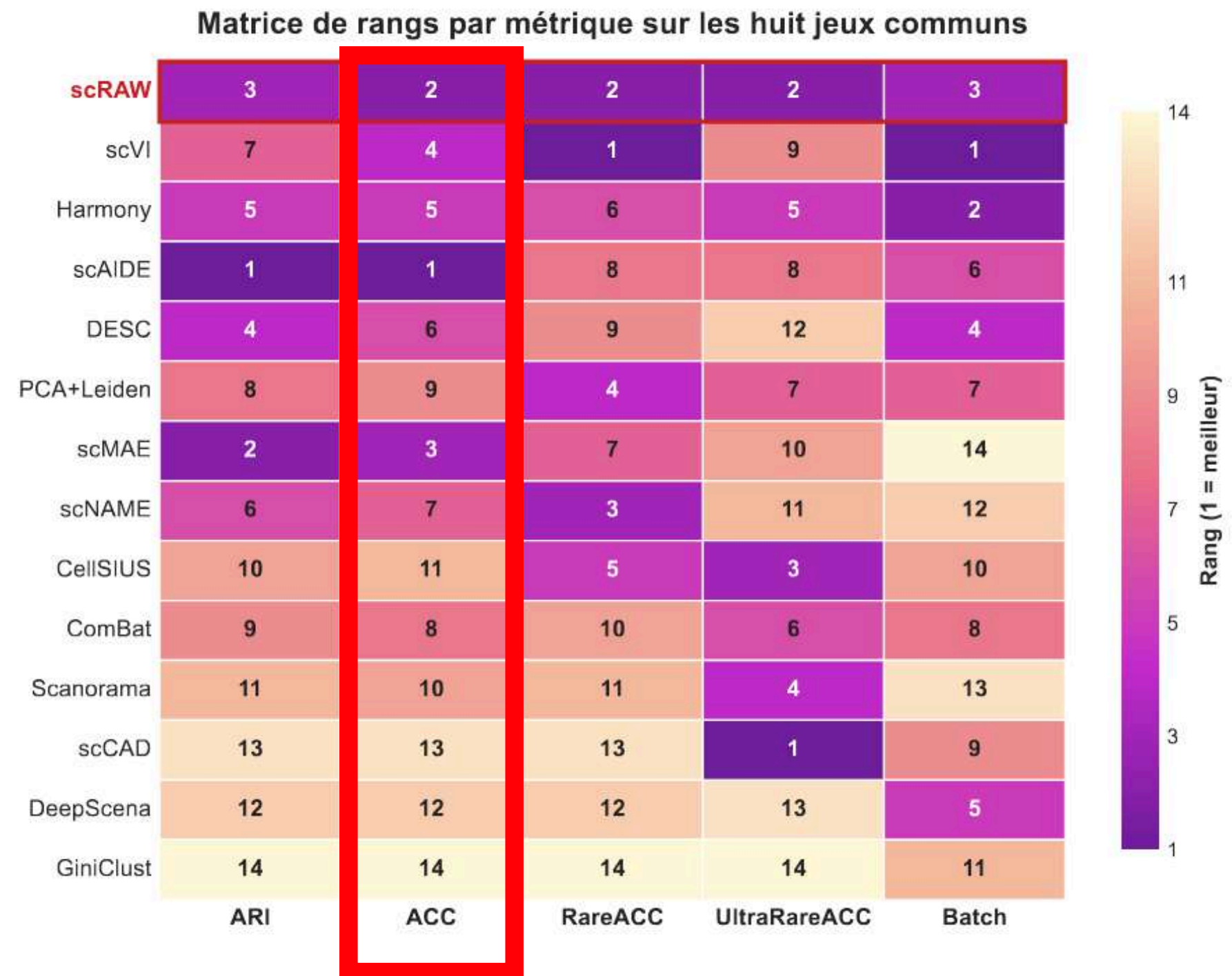
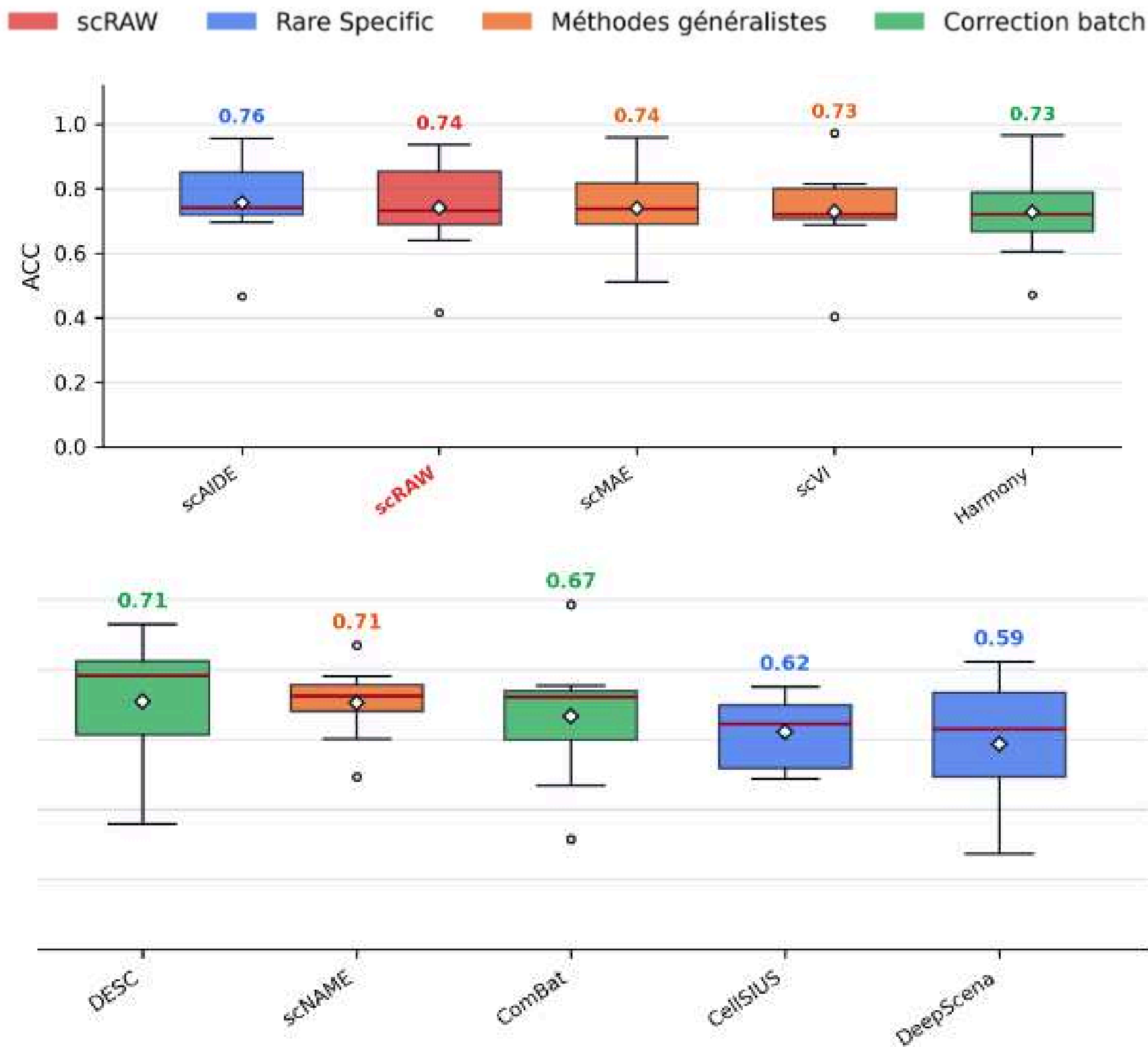
Dataset	Espèce	Cellules	Gènes	Batches	Rares (< 5%)	BATCH	Déséquilibre
BBAG094 Zeisel	Souris	3 006	19 972	10	4	0,499	0,543
BBAG094 spleen	Souris	9 552	23 341	2	3	0,022	0,644
Baron Human Pancreas	Humain	8 569	20 125	4	9	0,048	0,644
Human testis GSE112013	Humain	6 490	27 477	6	4	0,025	0,304
Kang PBMC	Humain	24 679	35 635	16	1	0,069	0,500
Macaque retina bipolar	Macaque	30 302	36 162	30	4	0,184	0,388
Paul15 bone marrow	Souris	2 730	3 451	1	9	0,000	0,410
Tabula Muris liver	Souris	2 859	22 966	1	5	0,000	0,578
Plage [Min - Max]		2 730–30 302	3 451–36 162	1–30	1–9	0,000–0,499	0,304–0,644

 Grande variété de datasets étudiées

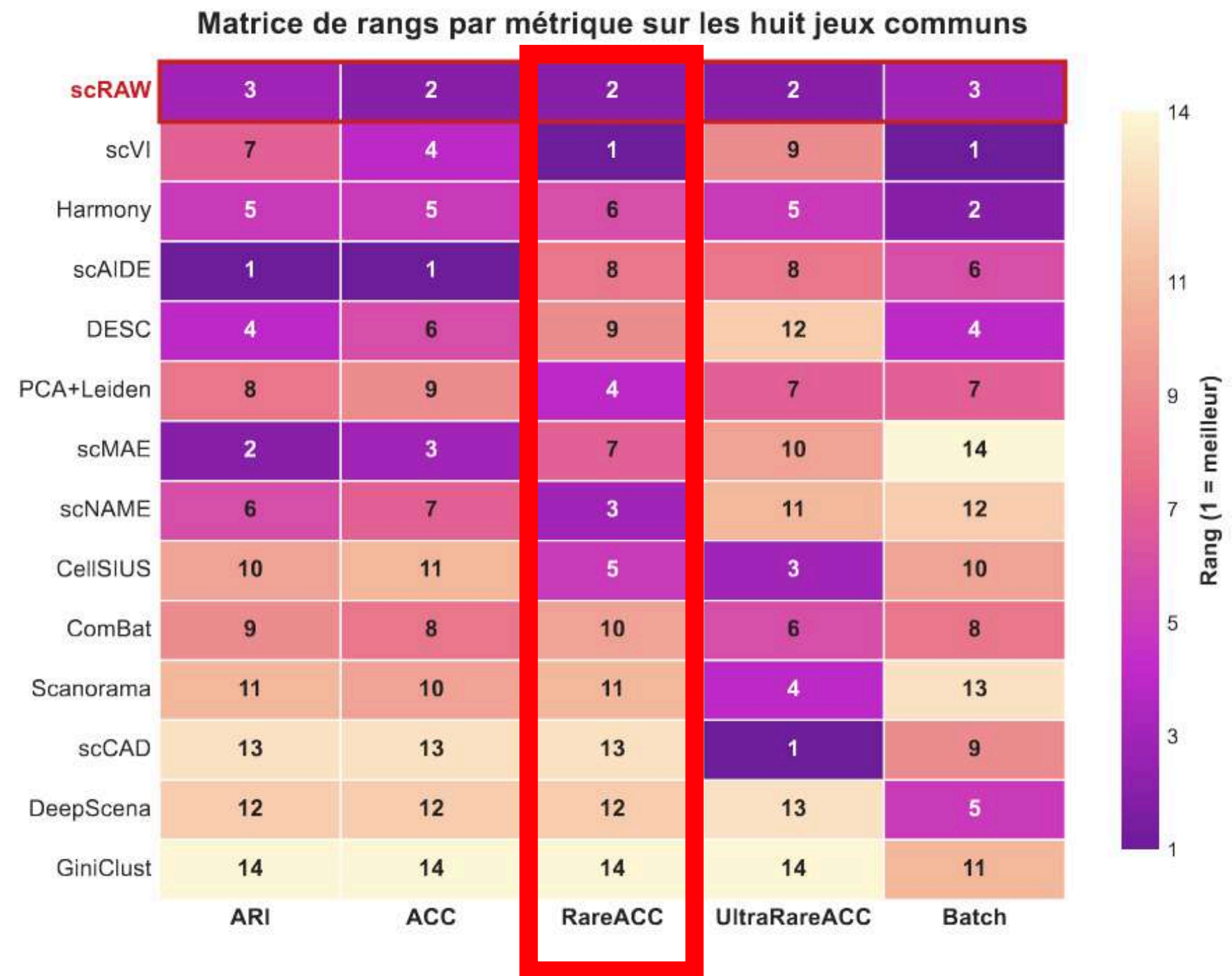
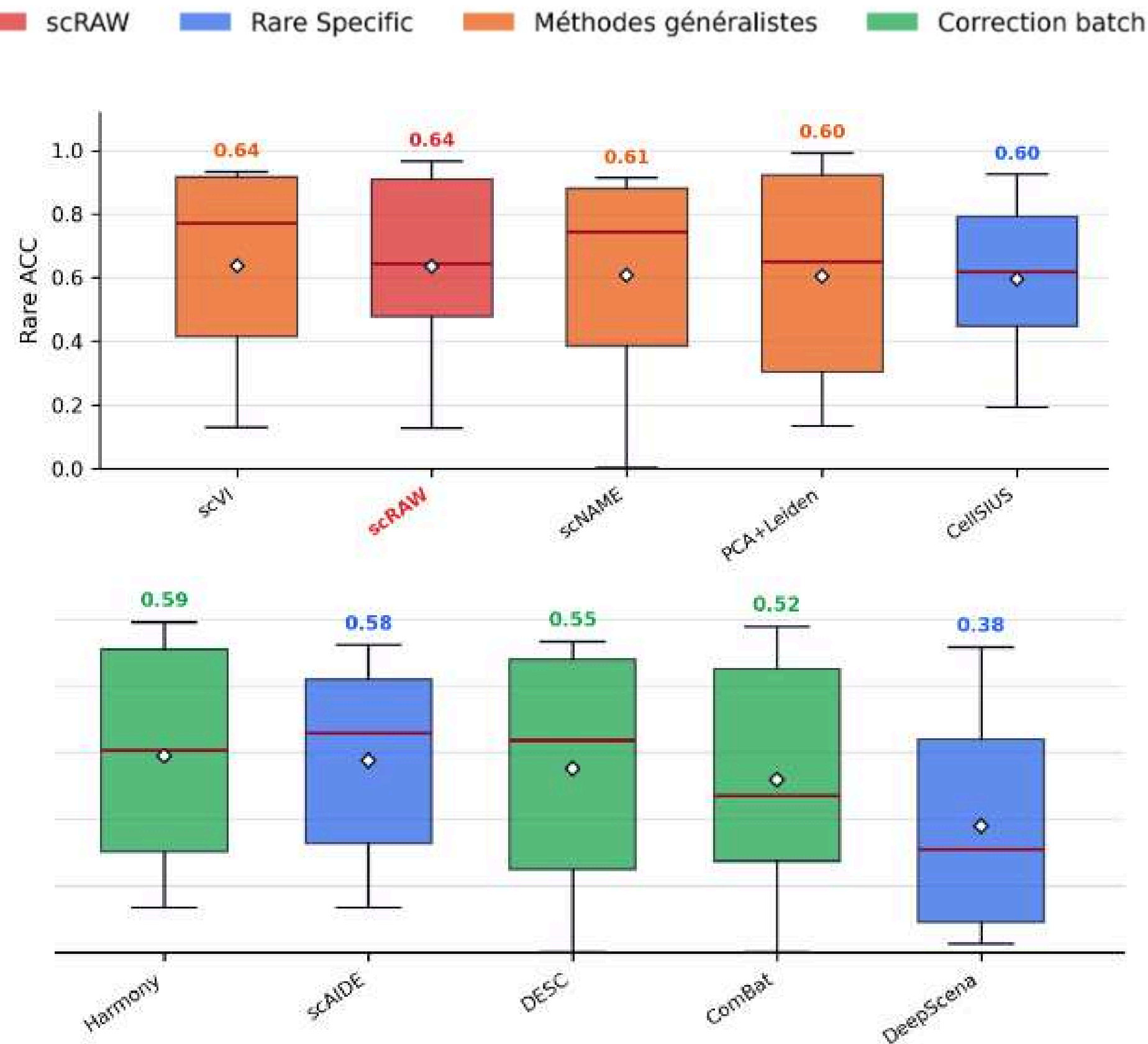
Résultats en clustering (ARI)



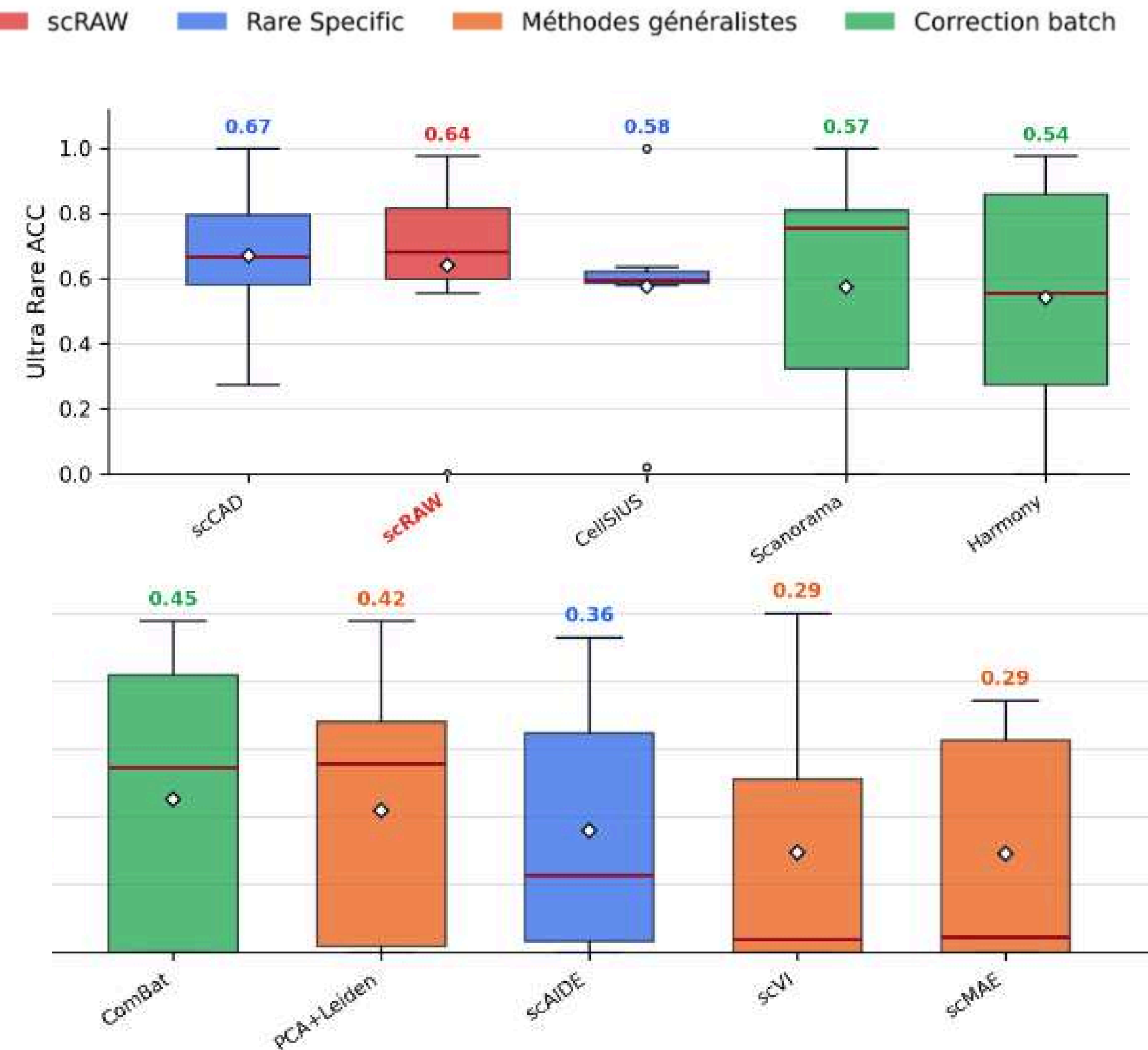
Résultats en accuracy (ACC)



Résultats en accuracy des cellules rares (RareACC)



Résultats en accuracy des cellules ultras-rares (UltraRareACC)

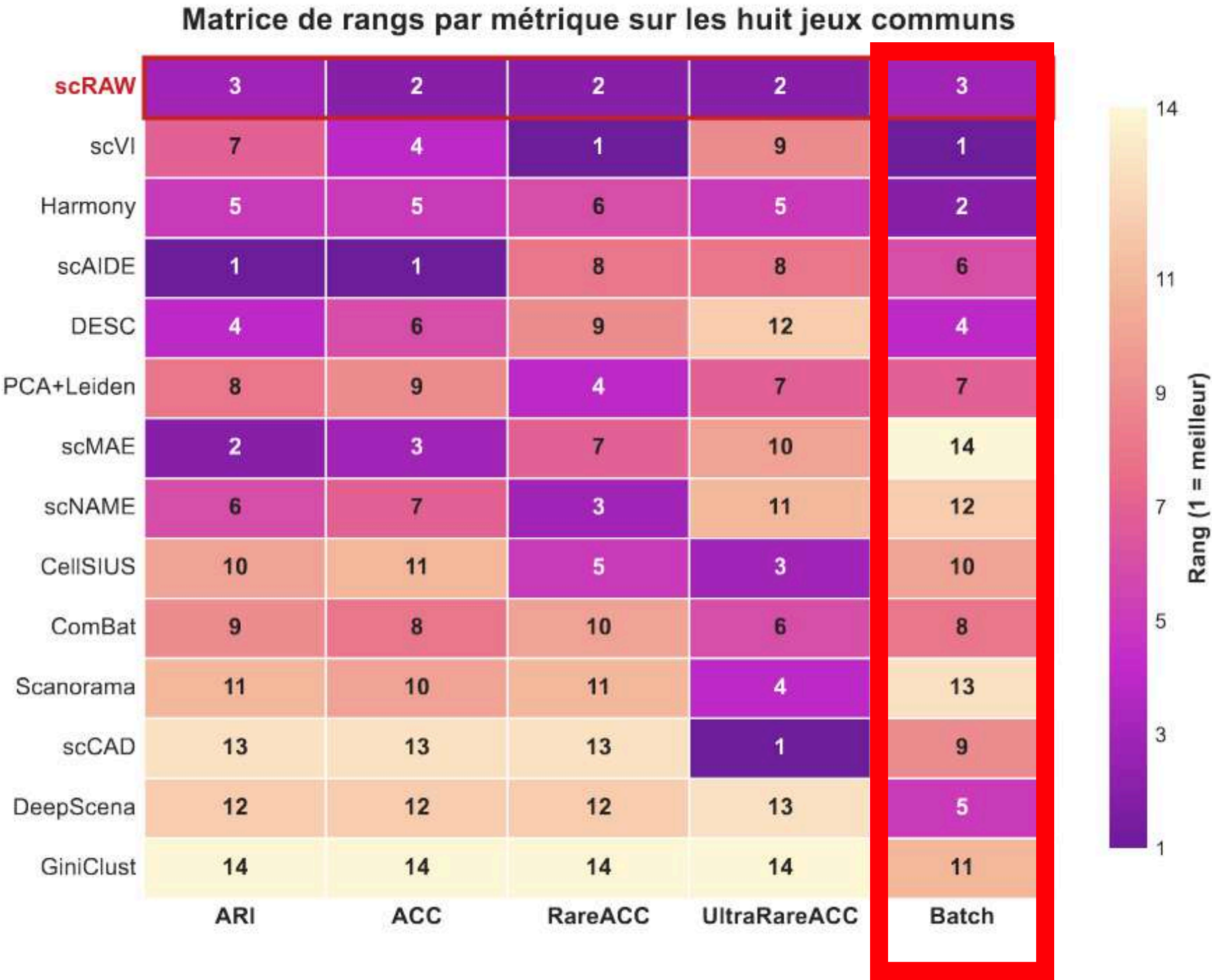
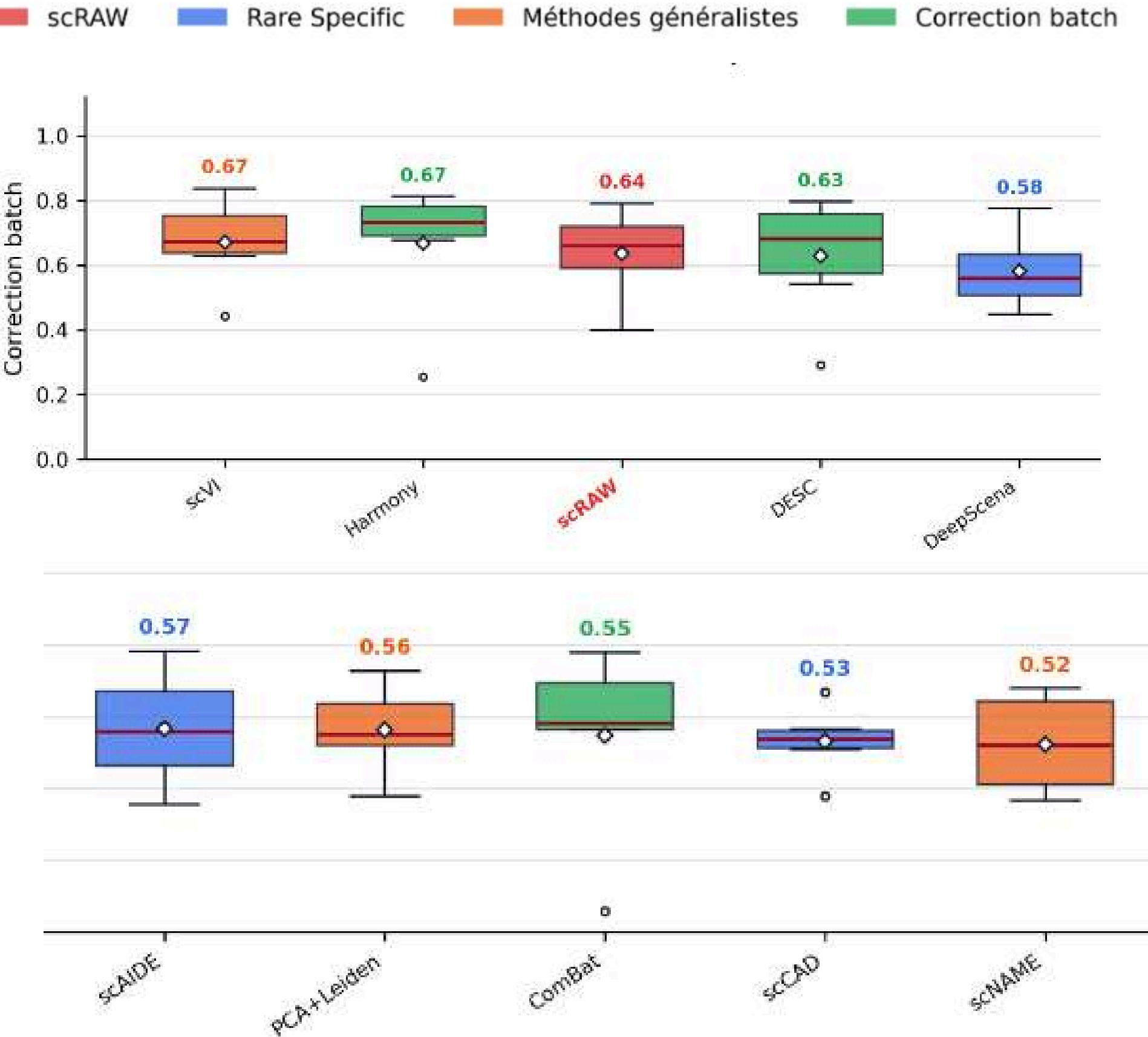


Matrice de rangs par métrique sur les huit jeux communs

	ARI	ACC	RareACC	UltraRareACC	Batch
scRAW	3	2	2	2	3
scVI	7	4	1	9	1
Harmony	5	5	6	5	2
scAIDE	1	1	8	8	6
DESC	4	6	9	12	4
PCA+Leiden	8	9	4	7	7
scMAE	2	3	7	10	14
scNAME	6	7	3	11	12
CellSIUS	10	11	5	3	10
ComBat	9	8	10	6	8
Scanorama	11	10	11	4	13
scCAD	13	13	13	1	9
DeepScena	12	12	12	13	5
GiniClust	14	14	14	14	11

Rang (1 = meilleur)

Résultats en correction du batch effect



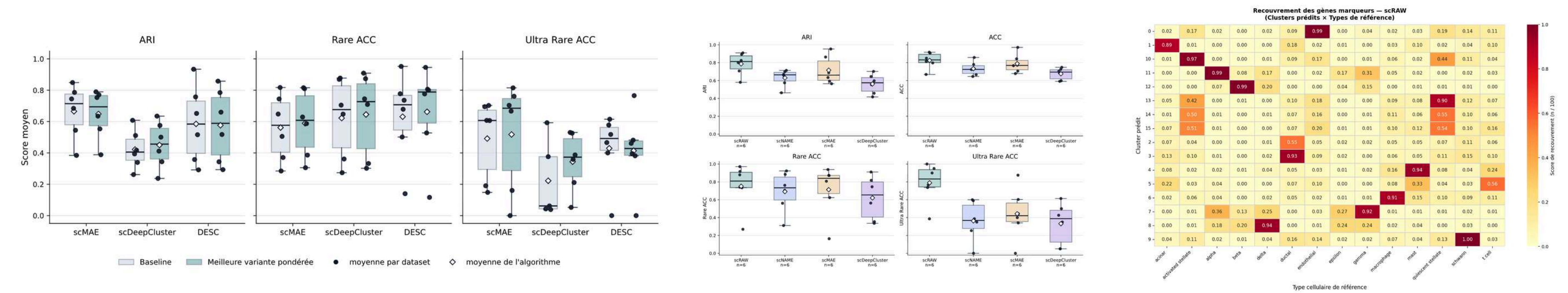
scRAW ne surpasse aucune méthode dans les métriques étudiées

scRAW présente **le meilleur compromis** entre clustering global, détection des cellules rares et correction de l'effet batch.

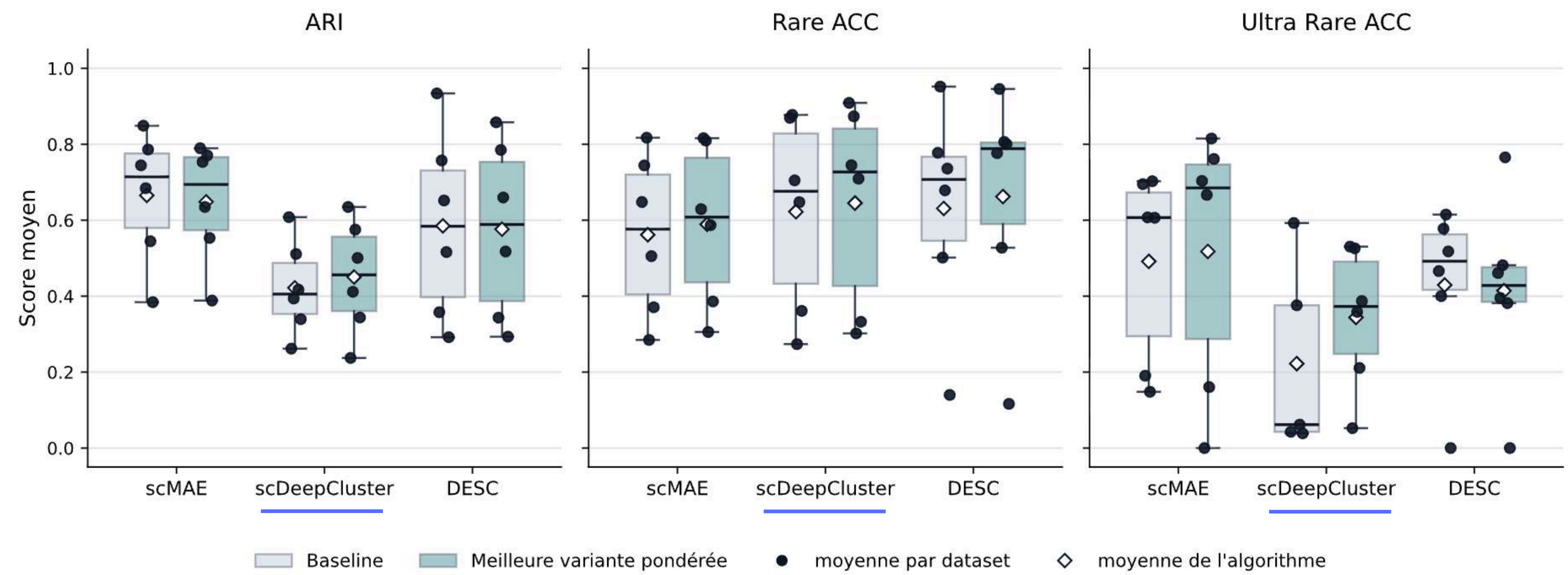
Limites : Plus de **répétitions**, plus de **datasets**, aller plus loin dans l'**intégration de jeu de données**

Expériences complémentaires issus de nos
travaux sur scRAW

- Intégration de la logique de pondération à d'autres méthodes de Deep Learning
- Test de la généralisation de notre modèle
- Interprétation biologiques

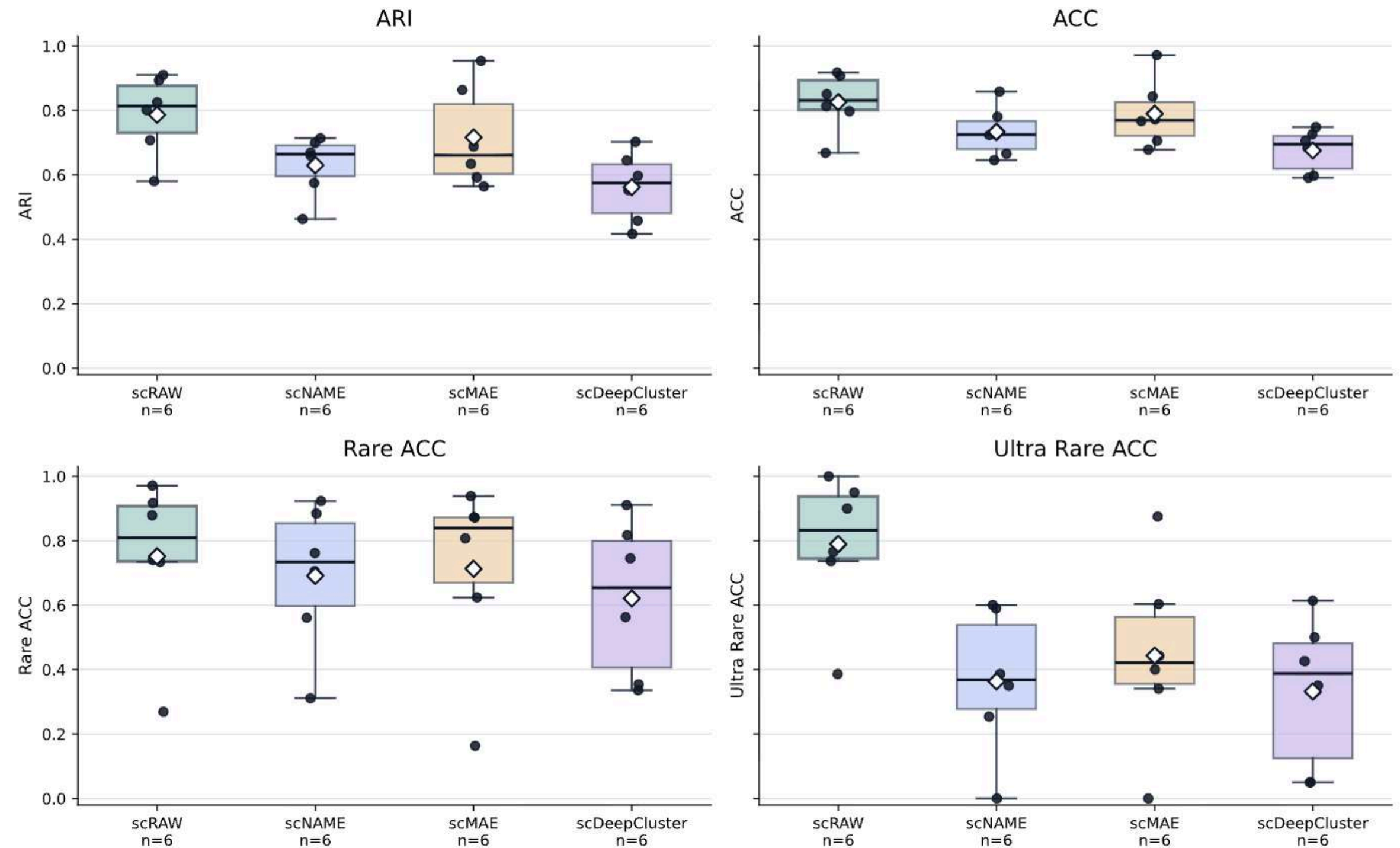


scRAW : Expériences complémentaires : Intégration de la logique de pondération



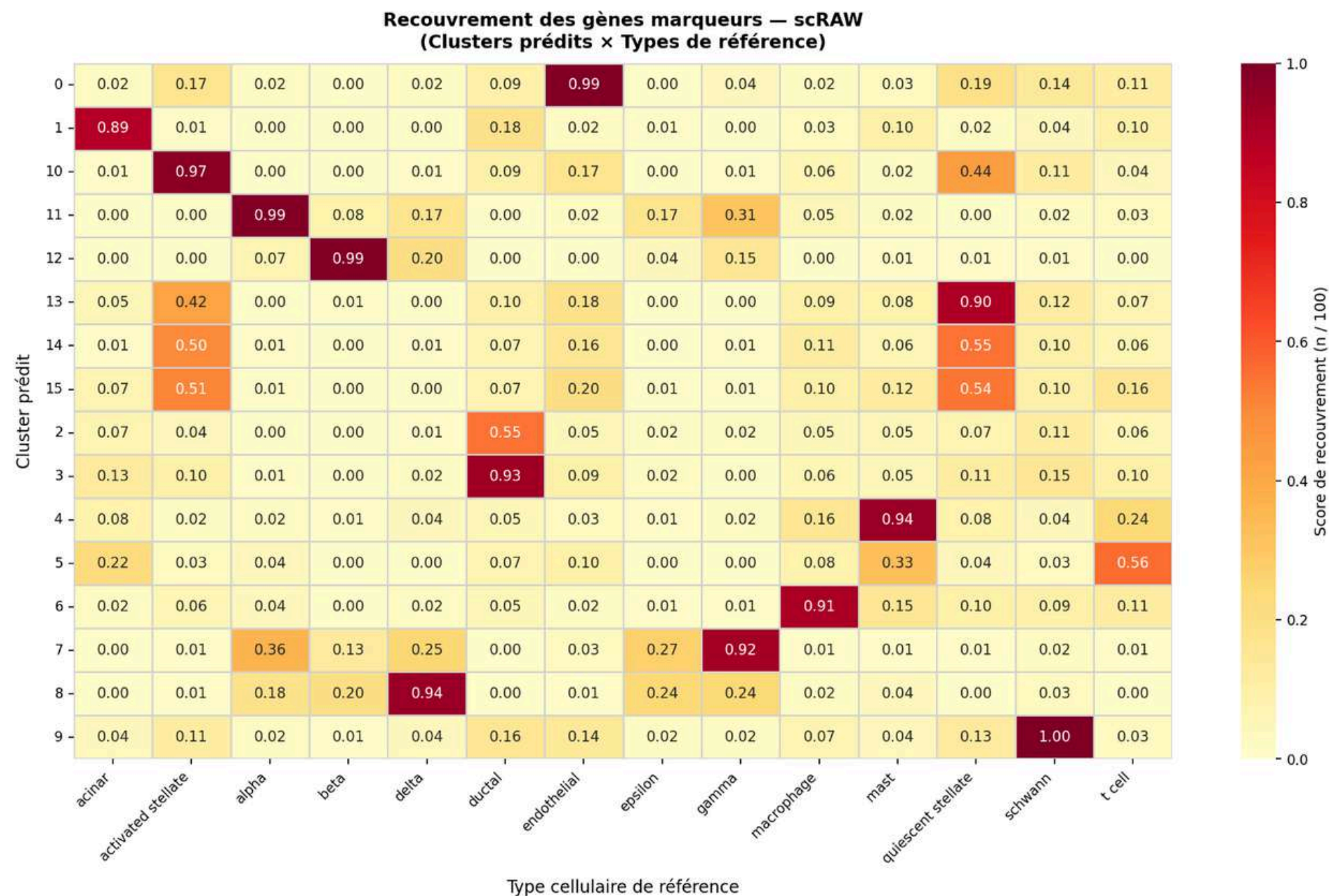
➡ Effectif que sur un certain type d'architecture

→ Nécessité d'adapter les méthodes

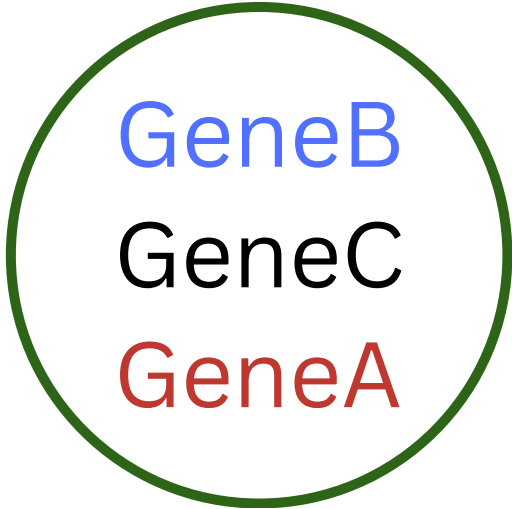


→ Piste intéressante notamment sur l'UltraRareACC

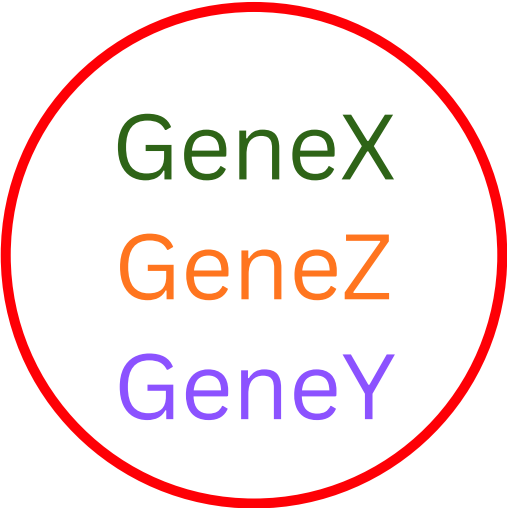
scRAW : Expériences complémentaires : Interprétation biologique



Prédictions

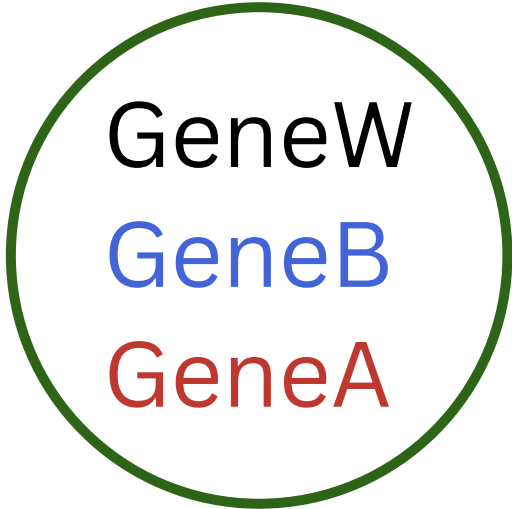


cluster 0

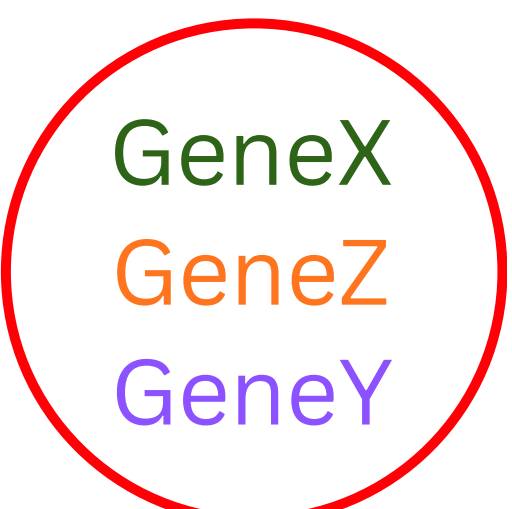


cluster 1

Vérité terrain



beta



alpha

$$\text{Recouvrement} = \frac{|\text{Marqueurs du Cluster} \cap \text{Marqueurs Réels}|}{100}$$

- Développement d'une **interface web** adaptée à des biologistes
- Développement d'un **algorithme de deep learning** pour le clustering non supervisé de données single-cell RNA-seq avec une loss pondérée sur les cellules rares
- Pistes de travaux encourageants sur la capacité de notre modèle à **généraliser** et à utiliser notre logique **d'intégration de loss** pour d'autres algorithmes.
- Nécessité d'approfondir les capacités sur **différents datasets** de notre modèle, les **expériences complémentaires, validations biologiques**